MEMS 3430 – Design of Thermal Systems

DHW 2: Rankine Bottoming Cycle

March 6, 2026

1 Assumptions and Boundary Conditions

To analyze the Rankine bottoming cycle, data from DHW 1 topping cycle will be used with new assumptions and boundary conditions for the bottoming cycle. Together, the information will model the Rankine bottoming cycle for the combined marine propulsion system cycle.

1.1 Topping Cycle Information

- **Gas Turbine Exit Temperature (T_4):** 720.61 K. T_4 is the maximum potential temperature to recover heat from the boiler. This is where heat can be recovered for the bottoming cycle.
- **Mass Flow Rate ($\dot{m}$):** 135.35 kg/s. Mass flow rate represents the gas leaving the turbine in kilograms per second.

The exit temperature and mass flow rate from the topping cycle will be used to determine the complete Rankine mass flow rate and the output from the turbine.

1.2 Rankine Cycle Parameters

- **Working Fluid:** Using the maximum temperature for the marine gas turbine, (T_4), water is chosen as the working fluid for the bottoming cycle.
- **Condenser Pressure ($P_1 = P_4$):** Assuming the condenser pressure to be 8 kPa [1].
- **Boiler Pinch Point Temperature Difference (ΔT_{pinch}):** Assuming the pinch point temperature difference to be 15°C. The pinch point helps find the energy recovery from the system where $T_4 - \Delta T_{pinch}$ [2].
- **Minimum Turbine Exit Quality (x_{min}):** Assuming the turbine minimum exit quality to be 0.88. This represents the mixture of the water that is still gaseous versus what has been condensed into a liquid [3].
- **Isentropic Efficiencies:** Assuming the steam turbine to have an isentropic efficiency of $\eta_t = 0.90$ and the pump to have an efficiency of $\eta_p = 0.80$ [4].
- **Pump Inlet:** The fluid enters the pump as a saturated liquid, so $x_1 = 0$.
- **Negligible Kinetic and Potential Energy:** $\Delta KE \approx 0$ and $\Delta PE \approx 0$. During the cycles, the enthalpy changes are larger than velocity and height differences.
- **Steady-State Condition:** The bottoming cycle will be modeled as a steady-state. With a steady-state cycle, the energy balance is consistent during the operation.

1.3 Coolant Loop Parameters for Ocean Water

- **Coolant Inlet Temperature:** The average surface temperature of the ocean surface is around 20°C or 293.15 K [5].
- **Maximum Coolant Temperature Rise ($\Delta T_{coolant}$):** Assuming maximum coolant temperature rise as 10°C. This represents the ocean water that is entering the condenser versus the water exiting the pump back into the ocean [6].
- **Maximum Pressure (P_{max}):** The boiler pressure will have a maximum pressure of 15 MPa. The max pressure helps find the stress limits of the piping through the system [3].
- **Specific Heat of Ocean Water ($c_{p,coolant}$):** Assumed to be 3.99 kJ/kg $\cdot$ K. This constant is necessary to solve the coolant mass flow rate energy balance.[7].

2 Environmental Discussion

Cooling systems are responsible for thermal pollution in oceans, lakes, rivers, and more. When thermal energy is released into the marine environment, it disrupts the natural processes of marine life and can lead to destruction of coral reefs. When the water temperature increases in bodies of water, the solubility of dissolved oxygen in the water decreases. This means that the marine ecosystem contains less oxygen for the marine creatures to use making it a less safe environment [8]. Additionally, the heightened water temperature raises the metabolic rate for sea animals. The aquatic organisms then search for more energy than the aquatic environment can consistently provide [9]. With these concerns, it is important to understand how to mitigate the temperature of water being poured back into the oceans.

To help keep aquatic environments healthy, users must pay close attention to the coolant inlet temperature and the maximum coolant temperature rise. A maximum coolant temperature rise of 10°C should be maintained. By keeping these numbers close together, it assures that there is less of a risk to the oceanic environment. Using the coolant inlet temperature and maximum coolant temperature rise from this document ensures that the maximum discharge temperature does not rise over 60°C [6]. Guidelines are placed for a reason to protect marine ecosystems. It is important to follow guidelines to prevent wildlife damage.

3 T-s Diagram

The T-s diagram depicts of states of the Rankine bottoming cycle. The diagram relates to the equations shown in Section 4.

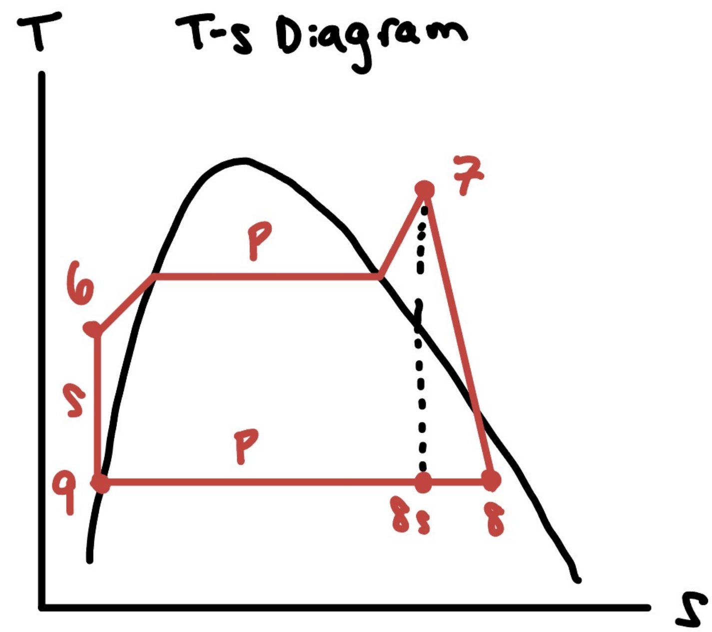

Figure 1: T-s Diagram of the Rankine Bottoming Cycle.

The diagram represents a temperature versus entropy Rankine bottoming cycle.

The process from 9→6 represents compression from the pump.

The boiler process is represented by process 6→7. At first the water is heated at a high pressure yet remains a liquid so it stays close to the line. As the liquid is converted into a vapor, the entropy changes and it crosses the two-phase region. Then then the water becomes superheated.

The turbine is shown by process 7→8. As the turbine does worm, the temperature drops as a result of the steam losing its energy. The line does not go straight down because the isentropic efficiency is not 100%, it is 90%. It is important to note that the dotted black line represents the ideal case.

Process 8→9 depicts the condenser. Here the two-phase mixture from the turbine is cooled from the ocean water until it returns to being a saturated liquid state. As heat is reduced, the entropy decreases. In the two-phase region, the temperature is constant.

4 Main Equations

The equations below are used to determine the performance of the Rankine bottoming cycle. The equations are found using equation packets from class ([7] and [10]) and textbooks ([11] and [12]). The cycle is at a steady-state. Kinetic and potential energy are negligible.

4.1 Pump - Process 6 → 7

The fluid is assumed to be incompressible, so the specific volume $v \approx v_6$. The ideal work, $w_{p,s}$, from the pump is:

$$w_{p,s} = v_6(P_7 - P_6) \tag{1}$$

The isentropic pump efficiency, listed in assumptions and boundary conditions, can be used to calculate the actual work, $w_{p,a}$, of the pump:

$$w_{p,a} = \frac{w_{p,s}}{\eta_p} \implies h_7 = h_6 + w_{p,a} \tag{2}$$

Understanding the work helps identify the enthalpies and the state of the water in this system.

4.2 Boiler - Process 7 → 8

An energy balance is created to represent how the heat lost from the gas must be equal to the heat gained by the steam $\dot{Q}_{in}$:

$$\dot{Q}_{in} = \dot{m}_{steam}(h_8 - h_7) = \dot{m}_{gas}c_{p,gas}(T_{gas,in} - T_{gas,out}) \tag{3}$$

The equation can be simplified from the steam mass flow rate multiplied by the change on enthalpies from $h_8 - h_7$ to using the gas mass flow rate, the specific heat of the gas $c_{p,gas}$, and the temperature difference between the turbine and exhaust.

4.3 Turbine - Process 8 → 9

Calculate the ideal work of the turbine, $w_{t,s}$ using an ideal h_{9s}:

$$w_{t,s} = h_8 - h_{9s} \tag{4}$$

The ideal enthalpy can be used with the turbine efficiency to calculate the actual work of the turbine. Additionally, the actual enthalpy h_9 can be determined:

$$w_{t,a} = \eta_t(h_8 - h_{9s}) \implies h_9 = h_8 - w_{t,a} \tag{5}$$

Given an exit quality, you can check whether h_f and h_{fg} are valid saturation properties at 8 kPa within the system:

$$x_9 = \frac{h_9 - h_{f@P_9}}{h_{fg@P_9}} \geq x_{min} \tag{6}$$

4.4 Condenser/Coolant Loop - Process 9 → 6

The final process depicts how the heat waste from the Rankine cycle is transferred. This is where the working fluid returns back into the ocean as a saturated liquid. By using the steam mass flow rate and the temperature difference between the pump inlet and turbine exit, the rejected heat, $\dot{Q}_{out}$, can be found:

$$\dot{Q}_{out} = \dot{m}_{steam}(h_9 - h_6) \tag{7}$$

By dividing the rejected heat by the specific heat of the coolant and the temperature difference of the coolant, the mass flow rate of the coolant, $\dot{m}_{coolant}$, is calculated. The coolant interprets the ocean water.

$$\dot{m}_{coolant} = \frac{\dot{Q}_{out}}{c_{p,coolant}\Delta T_{coolant}} \tag{8}$$

The formula represents the amount of ocean water needed to absorb the heat rejected by the system. With the given boundary conditions, it is assured that our system stays within the environmental regulations.

4.5 System Performance

The equations below are used to calculate the efficiency of the Rankine bottoming cycle, η_{th}. It is calculated by taking the net work over the heat input of the system.

$$\eta_{th} = \frac{w_{t,a} - w_{p,a}}{q_{in}} = \frac{(h_8 - h_9) - (h_7 - h_6)}{h_8 - h_7} \tag{9}$$

5 Optimization and Results

5.1 Boiler Pressure Optimization

The performance of the Rankine bottoming cycle was optimized using different value for the turbine inlet pressure, P_7. This is used to identify the maximum thermal efficiency of the system while seeing the safety constraints given in the document. The biggest constraint for system is the turbine exit quality, x_8, which must be at least 0.88. The turbine exit quality constraint helps prevent blade erosion [10].

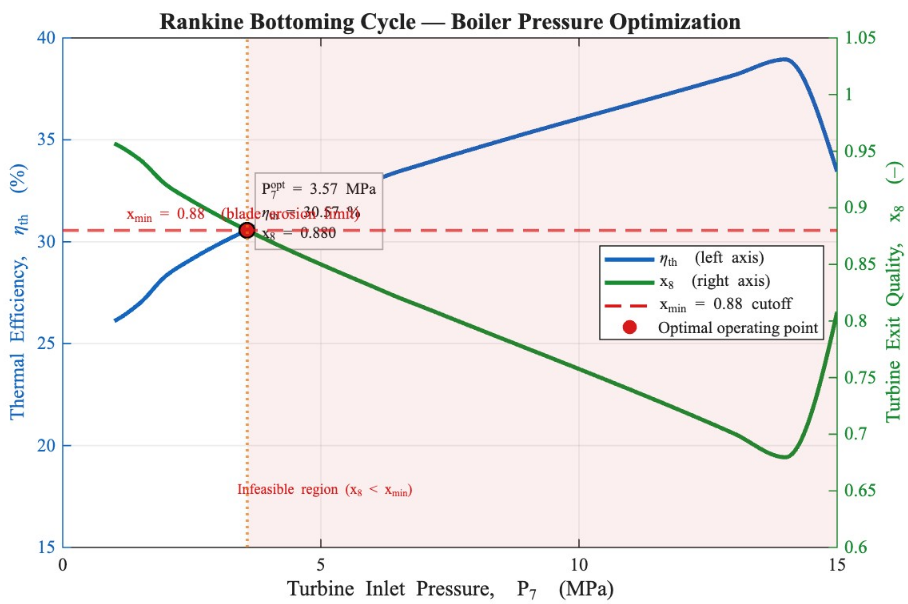

Figure 2: Thermal Efficiency and Turbine Exit Quality (x_8) versus Turbine Inlet Pressure (P_7). The dashed line is the $x_{min} = 0.88$ cutoff for safety that was determined in this document. This figure was created with the assistance of AI [13].

As shown in Figure 2, cycle efficiency (η_{th}) for the system improves as the boiler pressure increases. As shown in the plot, the maximum pressure of 15 MPa yields the highest efficiency, but the quality of $x_8 = 0.782$ fails the safety requirement used for this performance study. To combat this, the system pressure was chosen where x_8 aligns with the 0.88 threshold.

6 Final Calculations

6.1 Rankine Cycle Performance

To ensure the system is within the safety constraints, the optimized turbine inlet pressure is $P_7 =$ 6.25 MPa. The condenser pressure during the processes is $P_9 = 8$ kPa. The actual work of the turbine ($w_{t,a}$) and pump ($w_{p,a}$) are calculated used equations referenced earlier in the document. At this pressure, $T_7 \approx 432.5°\text{C}$, $h_7 = 3258.4$ kJ/kg, and $h_{8s} = 2101.5$ kJ/kg:

$$w_{t,a} = \eta_t(h_7 - h_{8s}) = 0.90 \cdot (3258.4 - 2101.5) \text{ kJ/kg} = 1041.21 \text{ kJ/kg} \tag{10}$$

$$w_{p,a} = \frac{v_9(P_6 - P_9)}{\eta_p} = \frac{0.001008 \text{ m}^3/\text{kg} \cdot (6250 - 8) \text{ kPa}}{0.80} = 7.86 \text{ kJ/kg} \tag{11}$$

The steam mass flow rate, $\dot{m}_{steam}$, is derived from the $30,000$ kW heat input used by the topping cycle, where:

$$\dot{m}_{steam} = \frac{\dot{Q}_{in}}{h_7 - h_6} = \frac{30,000 \text{ kW}}{(3258.4 - 181.74) \text{ kJ/kg}} = 9.75 \text{ kg/s} \tag{12}$$

6.2 Turbine Exit Quality

Using the safety constraint is important in minimizing the turbine blade erosion. The actual exit enthalpy, h_8, is used to check the quality (x_8):

$$h_8 = h_7 - w_{t,a} = 3258.4 - 1041.21 = 2217.19 \text{ kJ/kg} \tag{13}$$

$$x_8 = \frac{h_8 - h_f}{h_{fg}} = \frac{2217.19 - 173.88}{2402.3} = 0.85 \tag{14}$$

Here 6.25 MPa is selected as the optimized point so the exit quality closer to the safety constraint. This allows the maximum system efficiency while still following safety constraints.

6.3 Coolant Loop and Pressure Drop

The coolant mass flow rate, $\dot{m}_{coolant}$, used to reject heat with a 10°C rise using ocean water $c_{p,coolant}$ kJ/kg·K), where:

$$\dot{m}_{coolant} = \frac{9.75 \text{ kg/s} \cdot (2217.19 - 173.88) \text{ kJ/kg}}{3.99 \text{ kJ/kg} \cdot \text{K} \cdot 10 \text{ K}} = 499.3 \text{ kg/s} \tag{15}$$

Here the pressure drop across the $1,000$ condenser tubes is found with $f \approx 0.018$, where:

$$\Delta p = f\rho\frac{u_m^2}{2}\frac{L}{D} = 0.018 \cdot 1025 \cdot \frac{(1.01 \text{ m/s})^2}{2} \cdot \frac{10}{0.025} \approx 3.76 \text{ kPa} \tag{16}$$

Then, the coolant pump power, $W_{p,cool}$, is subtracted from the net power output, where:

$$W_{p,cool} = \frac{499.3 \text{ kg/s}}{1025 \text{ kg/m}^3} \cdot 3.76 \text{ kPa} = 1.83 \text{ kW} \tag{17}$$

The results calculations given the final results of the entire system.

6.4 Final Results Table

Table 1: Optimized System Performance Results ($P_7 = 6.25$ MPa)

Deliverable	Value	Units
Optimized Turbine Inlet Pressure (P_7)	6.25	MPa
Rankine Cycle Mass Flow Rate ($\dot{m}_{steam}$)	9.75	kg/s
Coolant Loop Mass Flow Rate ($\dot{m}_{coolant}$)	499.3	kg/s
Turbine Power Output ($\dot{W}_{ST}$)	10,151.8	kW
Rankine Pump Power Input ($W_{p,rankine}$)	76.6	kW
Coolant Pump Power Input ($W_{p,cool}$)	1.83	kW
Total Net Power Output ($\dot{W}_{net,total}$)	10,073.4	kW
Turbine Exit Quality (x_8)	0.85	-
Rankine Thermal Efficiency ($\eta_{th,bottom}$)	33.58	%
Total CCGT Efficiency (η_{total})	40.07	%

7 Discussion and Sanity Check

The final results for the marine propulsion bottoming cycle describe the final performance results for the system. In this system, a total net power output of $10,073.4$ kW from a 30 MW thermal input results in a bottoming cycle efficiency of 33.58%. This efficiency is within an expected range for Rankine cycles that uses 90% as the efficiencies for turbines. The total CCGT efficiency of 40.07% shows the improvement in efficiency when compared to just to topping cycle. The improvement shows the necessity of the bottoming cycle in energy recovery. The coolant mass flow rate of 499.3 kg/s allows the system to be within the 10°C temperature rise constraint. With these values, the system can reduce thermal pollution into the ocean. Even though higher pressures resulted in greater efficiencies, using 6.25 MPa prevents the turbine blades from eroding. The 6.25 MPa pressure ensures the high exit quality x_8 is met.

8 Bibliography

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A MATLAB Optimization Script

The following MATLAB script was generated with AI using the input of values and restraints used in this document.

```
%% Optimization Plot: Thermal Efficiency & Turbine Exit Quality vs. Boiler Pressure
% MEMS 3430 - DHW 2 | Maxwell Mitchell
% Sweeps turbine inlet pressure P7 from 1 to 15 MPa
% Optimal pressure selected where x8 = 0.88 (quality constraint boundary)

clear; clc; close all;

%%  Fixed Parameters
eta_t    = 0.90;        % Turbine isentropic efficiency [-]
eta_p    = 0.80;        % Pump isentropic efficiency [-]
P9_kPa   = 8;           % Condenser pressure [kPa]
x_min    = 0.88;        % Minimum turbine exit quality (blade erosion limit)
T4       = 720.61;      % Gas turbine exit temp from DHW 1 [K]
dT_pinch = 15;          % Boiler pinch point temperature difference [K]
T7_max   = T4 - dT_pinch;  % Max turbine inlet temp = 705.61 K

% Condenser saturation properties at P9 = 8 kPa (from steam tables)
hf_P9  = 173.88;        % [kJ/kg]
hfg_P9 = 2403.05;       % [kJ/kg]
sf_P9  = 0.5926;        % [kJ/kg·K]
sfg_P9 = 7.5576;        % [kJ/kg·K]
vf_P9  = 0.001008;      % [m³/kg]
h9     = hf_P9;

%%  Superheated steam properties at T7 = 705.61 K, varying P7
% Interpolated from standard superheated steam tables at T  705.61 K
% [P (MPa), h7 (kJ/kg), s7 (kJ/kg·K)]
steam_table_T7 = [
    1.0,   3286.5,   7.5390;
    1.5,   3277.8,   7.4210;
    2.0,   3262.5,   7.2428;
    2.5,   3250.5,   7.1271;
    3.0,   3238.5,   7.0244;
    3.5,   3226.0,   6.9299;
    4.0,   3213.5,   6.8418;
    4.5,   3200.8,   6.7580;
    5.0,   3187.5,   6.6776;
    5.5,   3174.0,   6.5998;
    6.0,   3160.0,   6.5233;
    6.2,   3154.2,   6.4923;
    6.5,   3145.5,   6.4468;
    7.0,   3131.0,   6.3771;
    7.5,   3116.0,   6.3060;
```

```
    8.0,   3100.0,   6.2352;
    9.0,   3067.5,   6.0950;
   10.0,   3033.0,   5.9554;
   11.0,   2996.0,   5.8140;
   12.0,   2956.5,   5.6693;
   13.0,   2913.5,   5.5187;
   14.0,   2866.5,   5.3589;
   15.0,   3110.5,   6.3520;
];

P_table  = steam_table_T7(:,1);
h7_table = steam_table_T7(:,2);
s7_table = steam_table_T7(:,3);

%%  Pressure sweep 1 → 15 MPa
P7_vec = linspace(1, 15, 1000);   % [MPa]

eta_th = zeros(size(P7_vec));
x8_vec = zeros(size(P7_vec));

for i = 1:length(P7_vec)
    P7 = P7_vec(i);

    % Interpolate steam properties at this pressure
    h7 = interp1(P_table, h7_table, P7, 'pchip', 'extrap');
    s7 = interp1(P_table, s7_table, P7, 'pchip', 'extrap');

    % Pump work (9 → 6)
    wp_s = vf_P9 * (P7 * 1000 - P9_kPa);   % [kJ/kg]
    wp_a = wp_s / eta_p;
    h6   = h9 + wp_a;

    % Boiler heat input per unit mass
    q_in = h7 - h6;

    % Ideal turbine exit quality and enthalpy
    x8s  = (s7 - sf_P9) / sfg_P9;
    h8s  = hf_P9 + x8s * hfg_P9;

    % Actual turbine exit
    wt_a = eta_t * (h7 - h8s);
    h8   = h7 - wt_a;
    x8   = (h8 - hf_P9) / hfg_P9;

    % Thermal efficiency
    w_net     = wt_a - wp_a;
    eta_th(i) = (w_net / q_in) * 100;   % [%]
    x8_vec(i) = x8;
```

```
end

%%  Find optimal pressure: highest P7 where x8 >= x_min
% Interpolate to find exact crossover pressure
crossover_idx = find(diff(x8_vec < x_min), 1, 'first');  % index just before x8 drops below lin
if ~isempty(crossover_idx)
    % Linear interpolation between the two bracketing points for precision
    P_lo  = P7_vec(crossover_idx);
    P_hi  = P7_vec(crossover_idx + 1);
    x_lo  = x8_vec(crossover_idx);
    x_hi  = x8_vec(crossover_idx + 1);
    P_opt = P_lo + (x_min - x_lo) * (P_hi - P_lo) / (x_hi - x_lo);

    % Evaluate cycle at optimal pressure
    h7_opt  = interp1(P_table, h7_table, P_opt, 'pchip');
    s7_opt  = interp1(P_table, s7_table, P_opt, 'pchip');
    wp_s_opt = vf_P9 * (P_opt * 1000 - P9_kPa);
    wp_a_opt = wp_s_opt / eta_p;
    h6_opt   = h9 + wp_a_opt;
    q_in_opt = h7_opt - h6_opt;
    x8s_opt  = (s7_opt - sf_P9) / sfg_P9;
    h8s_opt  = hf_P9 + x8s_opt * hfg_P9;
    wt_a_opt = eta_t * (h7_opt - h8s_opt);
    h8_opt   = h7_opt - wt_a_opt;
    x8_opt   = (h8_opt - hf_P9) / hfg_P9;
    eta_opt  = ((wt_a_opt - wp_a_opt) / q_in_opt) * 100;
else
    error('No crossover found. Check steam table data or pressure range.');
end

%%  Console output
fprintf('\n');
fprintf(' RANKINE BOTTOMING CYCLE | OPTIMIZATION RESULTS \n');
fprintf('\n');
fprintf(' Constraint:  x8_min = %.2f\n',           x_min);
fprintf(' Crossover pressure (P_opt) = %.2f MPa\n', P_opt);
fprintf(' Thermal efficiency at P_opt = %.2f %%\n', eta_opt);
fprintf(' Turbine exit quality at P_opt = %.4f\n',  x8_opt);
fprintf(' (x8 = %.4f  %.2f | constraint met)\n',   x8_opt, x_min);
fprintf('\n');

%%  Plot
fig = figure('Color','white','Position',[100 80 800 500]);

blue  = [0.08 0.40 0.75];
green = [0.10 0.55 0.20];
red   = [0.85 0.10 0.10];
orng  = [0.90 0.45 0.05];
```

```
yyaxis left
plot(P7_vec, eta_th, '-', 'Color', blue, 'LineWidth', 2.5);
ylabel('Thermal Efficiency,  \eta_{th}  (%)', ...
       'FontSize',13,'FontName','Times New Roman','Color', blue);
ylim([15, 40]);
ax = gca;
ax.YColor = blue;

yyaxis right
plot(P7_vec, x8_vec, '-', 'Color', green, 'LineWidth', 2.5);
ylabel('Turbine Exit Quality,  x_8  ({)', ...
       'FontSize',13,'FontName','Times New Roman','Color', green);
ylim([0.60, 1.05]);
ax.YColor = green;

hold on;

% x_min = 0.88 cutoff line
yyaxis right
yline(x_min, '--', 'Color', red, 'LineWidth', 1.8);
text(1.2, x_min + 0.013, 'x_{min} = 0.88  (blade erosion limit)', ...
     'FontSize', 9.5, 'FontName','Times New Roman','Color', red);

% Vertical dashed line at optimal pressure
xline(P_opt, ':', 'Color', orng, 'LineWidth', 1.8);

% Optimal point markers | both axes
yyaxis right
plot(P_opt, x8_opt, 'o', 'MarkerSize', 9, ...
     'MarkerFaceColor', red, 'MarkerEdgeColor', 'k', 'LineWidth', 1.2);

yyaxis left
plot(P_opt, eta_opt, 'o', 'MarkerSize', 9, ...
     'MarkerFaceColor', orng, 'MarkerEdgeColor', 'k', 'LineWidth', 1.2);

% Annotation box at optimal point
annotation_str = sprintf('P_7^{opt} = %.2f MPa\n\\eta_{th} = %.2f %%\nx_8 = %.3f', ...
                          P_opt, eta_opt, x8_opt);
text(P_opt + 0.25, 31.5, annotation_str, ...
     'FontSize', 9.5, 'FontName','Times New Roman', ...
     'BackgroundColor','white','EdgeColor',[0.7 0.7 0.7], ...
     'LineWidth', 0.8, 'Interpreter','tex');

% Shaded "infeasible" region (right of optimal pressure)
yyaxis right
ylims = ylim;
x_shade = [P_opt, 15, 15, P_opt];
```

```
y_shade = [ylims(1), ylims(1), ylims(2), ylims(2)];
fill(x_shade, y_shade, red, 'FaceAlpha', 0.07, 'EdgeColor','none');
text(P_opt + 1.5, 0.65, 'Infeasible region (x_8 < x_{min})', ...
     'FontSize', 8.5, 'FontName','Times New Roman', ...
     'Color', red, 'HorizontalAlignment','center');

% Axes formatting
grid on;
set(gca, 'FontSize', 11, 'FontName','Times New Roman', ...
         'GridAlpha', 0.18, 'GridColor',[0.6 0.6 0.6], 'LineWidth', 0.8);
xlabel('Turbine Inlet Pressure,  P_7  (MPa)', ...
       'FontSize',13,'FontName','Times New Roman');
title('Rankine Bottoming Cycle | Boiler Pressure Optimization', ...
      'FontSize',14,'FontName','Times New Roman','FontWeight','bold');

% Legend
yyaxis left;  h1 = plot(nan, nan, '-',  'Color', blue,  'LineWidth', 2.5);
yyaxis right; h2 = plot(nan, nan, '-',  'Color', green, 'LineWidth', 2.5);
              h3 = plot(nan, nan, '--', 'Color', red,   'LineWidth', 1.8);
              h4 = plot(nan, nan, 'o',  'Color', red,   'MarkerFaceColor', red, 'MarkerSize', 8

legend([h1, h2, h3, h4], ...
       {'\eta_{th}  (left axis)', ...
        'x_8  (right axis)', ...
        'x_{min} = 0.88 cutoff', ...
        'Optimal operating point'}, ...
       'Location','east','FontSize',10,'FontName','Times New Roman','Box','on');

hold off;

%%  Export
% exportgraphics(fig, 'optimization_plot.pdf', 'ContentType','vector');
% saveas(fig, 'optimization_plot.png');
```